

Translation



In this interview, Chinese AI expert Mu-ming Poo argues that DeepSeek's accomplishments are an example of China's historic excellence in putting new technologies to wide practical use. Poo contends that China should create academic scholarship programs modeled after DeepSeek's company culture of small teams of young, ambitious researchers working largely without supervision. He also predicts that LLM- and brain-inspired approaches to AI will converge in the next five years as the humanoid robot industry takes off.

Title

Mu-ming Poo: The Scientists Most Needed Today are Those Who are “Driven to Catch Big Fish,” Not Those Who Only Care About “Fishing for the Joy of it”

视频|蒲慕明：目前最需要的科学家，不是只顾“钓鱼之乐”，而是“要钓大鱼”

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Mu-ming Poo: The Scientists Most Needed Today are Those Who are “Driven to Catch Big Fish,” Not Those Who Only Care About “Fishing for the Joy of it”

The five-year plan for the second phase of the 2030 Major S&T Project “Brain Science and Brain-Inspired Research” (the “China Brain Project”) is currently being formulated. Large language models (LLMs) seem to be more powerful than the human brain in many respects. Do we still need to study the brain? What insights has the DeepSeek “frenzy” given the Chinese scientific community? This *Liberation Daily* reporter sat down for an exclusive interview with an important leader in the field of brain science in China—Academician Mu-ming Poo (Pu Muming; 蒲慕明), Academic Director of the

Center for Excellence in Brain Science and Intelligence Technology (CEBSIT) of the Chinese Academy of Sciences (CAS).

[“Going from 1 to 100” at an unprecedented speed]

Liberation Daily: What is your evaluation of DeepSeek’s “emergence out of nowhere?”

Mu-ming Poo: 2025 will go down as one of the most memorable years for Chinese science in recent decades. The reasoning artificial intelligence (AI) model DeepSeek released, R1, is comparable in performance to the strongest model in the field, OpenAI-o1. It has cleverly used a mix of expert models, reinforcement learning, and distillation methods to achieve unexpected inference efficiency. And for the first time, it shows users the reasoning “chain of thought.” Surprisingly, the cost of pre-training the model was far lower than that of current top LLMs.

Of course, DeepSeek-R1 is not a groundbreaking new technology like the backpropagation algorithm or the Transformer (transformation model) architecture, which triggered the two AI revolutions of deep learning and LLMs, respectively. The backpropagation algorithm has also been recognized with the 2024 Nobel Prize in Physics. What I would like to emphasize, however, is that although DeepSeek-R1 is not an original breakthrough “from 0 to 1,” it is AI development “from 1 to 100” at an unprecedented speed. It clearly shows that development “from 1 to 100” may have a more significant impact than development “from 0 to 1.”

Liberation Daily: What thoughts does DeepSeek’s going “from 1 to 100” give rise to?

Mu-ming Poo: There are two points to reflect upon. First, China’s AI research institutions have received a lot of funding from the government, and the number of papers published in recent years has grown astonishingly, but truly influential discoveries or inventions in the AI field are still rare. DeepSeek is a small AI company in Hangzhou. Its head, Liang Wenfeng (梁文峰), doesn’t have an outstanding academic background. His papers are only published in ordinary “technical reports” and he is rarely invited to speak at AI academic conferences, yet this young team has done such influential work.

Second, there is a scaling law in the AI community, which holds that the larger the model and the greater the number of parameters, the higher the computing power and the higher the performance. Although there is evidence that the law is approaching its limits, it is still the mainstream view. DeepSeek has taken a different path—algorithm innovation. I have heard some AI researchers talk about how they wouldn’t dare do something so ambitious because the risk is too high. Most of the Chinese scientific community is now doing incremental “from 1 to 10” research, which is not “from 0 to

1” original breakthroughs, nor does it have “from 1 to 100” application value. However, it is easy to publish papers with this type of work, so gradually progressing, small-increment research is prevalent. DeepSeek’s goal is to explore how to achieve general purpose artificial intelligence,¹ the key issues of which are reasoning and generalization capabilities. They focus on the most important key issues, and explore solution paths under the constraints of limited resources. It is precisely because they have passion, and also have the courage to overcome important problems, that they have achieved development by leaps and bounds.

[The second part of “the Needham Question”]

Liberation Daily: “From 1 to 10” papers are prevalent. What is the deeper reason behind this?

Mu-ming Poo: The reason behind this is that our scientific research is influenced too much by European countries and the United States. We only want to follow them closely, and we rarely have the courage to overtake them. Take neuroscience as an example. Most of our research is of the “follow-up” type, and is not “groundbreaking.”

However, scientific research approaches in Europe and the United States are not necessarily suitable for us. I have compiled some statistics: From 1970 to the present, 25% of the Nobel Prizes in Medicine or Physiology have been directly related to neuroscience, but why, then, has not a single brain disease been cured? To develop a drug, from the discovery of the target to becoming a marketed drug, you must do “from 1 to 100” work. In my opinion, European and U.S. scientists are more inclined to do the most front-end “0 to 1” research that can win the Nobel Prize, and the “1 to 10” research that can produce high-level papers, as well as some so-called “proof of principle” laboratory results. They don’t really get involved in the “10 to 100” innovative research and development work that has practical applications on the back end.

Everyone knows the first “Needham Question,” but not the second. In fact, he condensed two important questions. The first question is: Why did modern science not emerge in Chinese civilization, but only develop in Europe? This question is widely

¹ Translator's note: The Chinese term 通用人工智能 can be translated into English either as "artificial general intelligence" (AGI) or "general purpose artificial intelligence" ("general purpose AI," for short). This translation opts for "general purpose AI" because when Chinese writers use the term 通用人工智能, they are usually referring to a broadly capable form of AI, rather than to AI that resembles human cognition, as AGI implies. For a more thorough discussion of this term, see Wm. C. Hannas, Huey-Meei Chang, Daniel H. Chou, and Brian Fleeger, "China's Advanced AI Research: Monitoring China's Paths to 'General' Artificial Intelligence," Center for Security and Emerging Technology, July 7, 2022, <https://cset.georgetown.edu/publication/chinas-advanced-ai-research/>, pp. 1-3.

known, and his answer was “bureaucratic feudalism.” The second question is: Why was Chinese civilization much more effective than Western civilization in applying human knowledge of nature to people’s practical needs from the 5th to the 13th century, and what unique contributions did it make to the world? This question is not well known, and his answer was “the tradition of pragmatism.”

Many of China’s scientific and technological achievements before the modern era originated based on practical applications, and were goal-oriented. Science and technology were closely integrated. For example, astronomy, mathematics, chemistry, biology, engineering, and maritime navigation technology were all directly related to people’s practical needs. I think this is a very good scientific tradition. It is in the same vein as basic research oriented towards practical applications, which is now encouraged.

I often draw an analogy: Scientific exploration is like going fishing at sea. The goal is very important: First you have to figure out whether you want to catch a big fish or a small fish. Because if the goal is to catch a big fish, you won’t head home once you’ve caught some small fish. Also, there is a fundamental difference between exploration that is about “fishing for the joy of it” and exploration that is about “being driven to catch big fish.” I believe that what our society needs most at the moment are scientists who are “driven to catch big fish,” not those who only care about “fishing for the joy of it.”

Liberation Daily: What has impressed you most deeply about DeepSeek’s innovative culture?

Mu-ming Poo: They encourage bottom-up, proactive innovation. As long as you have a good idea, you can put it forward, and then resources will be concentrated on it. This is very much related to the structure of “small teams.” Whether it was Apple, Microsoft, or Meta’s predecessor, Facebook, they all started out as small teams, and some were tinkering in their own garages. Small teams like DeepSeek are more efficient in terms of rapid closed-loop interaction, focusing on a set goal and coalescing around it, and the spirit of interconnection between team members. I hope that as DeepSeek continues to expand, the culture established during the early small-team days can continue to be maintained.

Most scientific research organizations in China are unable to achieve “bottom-up, proactive innovation.” Very little room is given to young people to pursue their own ideas.

Liberation Daily: How can the innovative talents of young researchers be unleashed?

Mu-ming Poo: I admire the Whitehead Fellows Program at the Massachusetts Institute

of Technology. For exceptionally gifted recent PhD graduates in biology, the program does not require them to first do postdoctoral research as is customary, but rather directly provides them three years of research funding and an independent research laboratory. It cultivates young scientists during their most creative period. The best example is Zhang Feng (张锋), who produced Nobel Prize-worthy results—the first application of CRISPR-Cas9 gene editing technology to mammalian and human cells—while he was a Whitehead Fellow.

Another example is Gerald Maurice Edelman. When he was a graduate student at the Rockefeller University in the United States, he had no fixed advisor. He conducted research independently, and went on to win the Nobel Prize for his discovery of the chemical structure of antibodies.

At my suggestion, the Center for Excellence in Brain Science and Intelligence Technology at the Chinese Academy of Sciences has launched a similar young researcher program: Three outstanding recent PhD graduates were hired as young researchers, and they have their own independent laboratories. After a three-year period, they can be directly promoted to researchers if they do outstanding work. So far, this initiative has been quite effective.

I even have another idea: Could the best students among recent graduates be selected and given two or three years of research funding and resources? There would be no need to assign advisors, but they could go to any “advisor team” they needed for advice. This kind of environment would especially encourage young people to pursue their own ideas. The saying “young people are fearless” (初生牛犊不怕虎) makes sense.

[Do we still need to study the brain to advance AI?]

Liberation Daily: The birth of AI was closely related to neuroscience, and artificial neuron models were directly inspired by biological neurons. But why do most AI researchers now pay little attention to the working principles of the brain?

Mu-ming Poo: After the advent of deep learning models, AI researchers who had previously been somewhat interested in brain science haven’t paid much attention to it, because they are so convenient and powerful. They rely more on mathematical optimization and large-scale data than directly imitating the mechanisms of the brain. The deep learning and reinforcement learning modes of artificial networks are different from those of the brain.

Liberation Daily: LLMs seem to be more powerful than the human brain in many ways. Do we still need to study the brain to advance AI?

Mu-ming Poo: An LLM is a piece of software. It is only when that software is

integrated into a physical system that it can truly interact with the environment. I believe that humanoid robots with high-level embodied AI (具身智能) will be the most active field in the next 5 to 10 years, but the key technology of how to combine LLMs with sensory and motion systems has not been overcome yet. For example, to make a humanoid robot with general purpose AI, we would need an LLM that directly interacts with the external world. The LLM can serve as the brain of the robot. It needs to be connected to a sensory system that perceives the external world and a motion system that outputs movement. By interacting with the external world through these two physical systems, the LLM can build a continuously evolving “world model” that can make distinctions based on experience, one that continuously models and understands the external world, and thereby carries out appropriate decision-making and planning.

The combination of large models and physical systems will bring about the next leap in AI, and brain science has a lot to offer. We need to pay more attention to studying how the brain achieves efficient calculation, as well as how human intelligence is embodied in a system that can effectively interact with the external world. For example, integrating multimodal sensory information is precisely what the human brain is good at. There is a “cocktail party phenomenon” where, even in a noisy environment, we can still focus our attention on the conversation of the person we are interested in while filtering out other irrelevant noise. This requires effective integration of vision and hearing. Through the brain’s instructions, the human motion system can activate hundreds of muscles in parallel and sequential patterns to perform smooth and precise movement. How would an LLM’s instructions activate such a motion system? Brain science can make key contributions.

Liberation Daily: The combination of large models and physical systems—will the second phase of the China Brain Project consider that?

Mu-ming Poo: Some experts have already made this suggestion. The focus of the China Brain Project is “brain science and brain-inspired research.” Humanoid robots with advanced embodied intelligence will be an excellent platform for combining brain science and brain-inspired intelligence technology.

Developing embodied intelligence in physical systems such as humanoid robots requires close interdisciplinary cooperation between software engineers, bioengineers, and neuroscientists. I look forward to seeing this in the next five years.